

Annex 31A

(normative)

MAC Control opcode assignments

Table 31A–1 shows the currently defined opcode values and interpretations:

Table 31A–1—MAC Control opcodes

Opcode (Hexadecimal)	MAC Control function	Specified in	Value/Comment	Timestamp ^a
00-00	Reserved			
00-01	PAUSE	Annex 31B	Requests that the recipient stops transmitting non-control frames for a period of time indicated by the parameters of this function.	No
00-02	GATE	Clause 64, Clause 77, Clause 103	Request that the recipient allows transmission of frames at a time, and for a period of time indicated by the parameters of this function.	Yes
00-03	REPORT	Clause 64, Clause 77, Clause 103	Notify the recipient of pending transmission requests as indicated by the parameters of this function.	Yes
00-04	REGISTER_REQ	Clause 64, Clause 77, Clause 103	Request that the station be recognized by the protocol as participating in a gated transmission procedure as indicated by the parameters of this function.	Yes
00-05	REGISTER	Clause 64, Clause 77, Clause 103	Notify the recipient that the station is recognized by the protocol as participating in a gated transmission procedure as indicated by the parameters of this function.	Yes
00-06	REGISTER_ACK	Clause 64, Clause 77, Clause 103	Notify the recipient that the station acknowledges participation in a gated transmission procedure.	Yes
00-07 through 00-11	Reserved			
00-12	GATE	144.3.6.1	Request that the recipient allows transmission of frames at a time, and for a period of time indicated by the parameters of this function.	Yes
00-13	REPORT	144.3.6.2	Notify the recipient of pending transmission requests as indicated by the parameters of this function.	Yes
00-14	REGISTER_REQ	144.3.6.3	Request that the station be recognized by the protocol as participating in a gated transmission procedure as indicated by the parameters of this function.	Yes

Table 31A–1—MAC Control opcodes (*continued*)

Opcode (Hexadecimal)	MAC Control function	Specified in	Value/Comment	Timestamp ^a
00-15	REGISTER	144.3.6.4	Notify the recipient that the station is recognized by the protocol as participating in a gated transmission procedure as indicated by the parameters of this function.	Yes
00-16	REGISTER_ACK	144.3.6.5	Notify the recipient that the station acknowledges participation in a gated transmission procedure.	Yes
00-17	DISCOVERY	144.3.6.6	Request that recipients attempt registration during the discovery window.	Yes
00-18	SYNC_PATTERN	144.3.6.7	Announces burst synchronization patterns to all unregistered ONUs, multiple/all registered ONUs, or individual registered ONUs.	Yes
00-19 through 00-1F	Reserved			
00-20	CC_REQUEST	144.4.3.1	Query or change the state of ONU channel(s).	No
00-21	CC_RESPONSE	144.4.3.2	Report current channel(s) state and action result code.	No
00-22 through 01-00	Reserved			
01-01	PFC	Annex 31D and IEEE Std 802.1Q	Requests that the recipient stops transmissions in the priorities indicated in the parameters of the function for a period of time also indicated in the parameters.	No
01-02 through FF-FD	Reserved			
FF-FE	EXTENSION	Annex 31C	This frame is used for Organization–Specific Extension. Upon reception of this message, the MAC Control generates MA_CONTROL.Indication informing the MAC Control Client to perform the relevant action.	No
FF-FF	Reserved			

^aThe value of the Timestamp field is generated by MAC Control and is not exposed through the client interface.

Opcodes are transmitted high-order octet first. Within each octet, bits are transmitted least-significant bit first. Reserved opcodes shall not be used by MAC Control sublayer entities.

Table 31A–2 through Table 31A–9 show the elements and semantics of the indication_operand_list for MA_CONTROL.indication primitives for each currently defined opcode value in Table 31A–1:

Table 31A–2—PAUSE MAC Control indications

PAUSE (opcode 0x0001)		
indication_operand_list element	Value	Interpretation
pause_status	paused	Indicates that the PAUSE function is inhibiting transmission of data frames by the MAC client.
	not_paused	Indicates that the PAUSE function is not inhibiting transmission of data frames by the MAC client.

Table 31A–3—GATE MAC Control indications

GATE (opcode 0x0002)		
indication_operand_list element	Value	Interpretation
start	32 bit	Time when transmission should be initiated.
length	16 bit	Interval of time during which transmission is allowed.
status	arrive	Indicates that a grant has been received for future activation.
	active	Indicates that the GATE function is allowing transmission of frames.
	deactive	Indicates that the GATE function is inhibiting transmission of frames.
force_report	true	The OLT expects the ONU to transmit a REPORT message during the transmission opportunity identified by start and length fields.
	false	The OLT does not request the ONU to transmit a REPORT message during the transmission opportunity identified by start and length fields.
discovery	true	This grant is reserved for use during discovery.
	false	This grant is not reserved for use by the discovery process, and is available for use by all traffic.
discoveryInformation ^a	16 bits	See Table 77–3 for the internal structure of the discoveryInformation field.

^aOnly present in 10G–EPON GATE MAC Control indication (Clause 77).

Table 31A–4—REPORT MAC Control indications

REPORT (opcode 0x0003)		
indication_operand_list element	Value	Interpretation
RTT	32 bit	Indicates the calculated round trip time for the station, as calculated following the REPORT message reception.
n	Integer	Indicates the number of queue status reports present
report_list	valid[8]	Indicates whether the corresponding status element is present when set to true.
	status[8]	Indicates amount of data waiting in the corresponding queue including the associated transmission overhead.

Table 31A–5—REGISTER_REQ MAC Control indications

REGISTER_REQ (opcode 0x0004)		
indication_operand_list element	Value	Interpretation
status	incoming	Indicates that a station is requesting recognition.
	retry	Indicates that the station should reattempt registration.
flags	register	Indicates that the station is requesting to register.
	deregister	Indicates that the station is requesting to deregister.
pending_grants	Integer	Indicates the maximum number of future GATES that can be stored by the GATE function.
RTT	32 bit	Indicates the calculated round trip time for the station, as calculated following the REGISTER_REQ message reception.
laserOnTime ^a	8 bits	Indicates the Laser On Time characteristic for the given ONU transmitter, expressed in the units of time_quanta.
laserOffTime ^a	8 bits	Indicates the Laser Off Time characteristic for the given ONU transmitter, expressed in the units of time_quanta.
discoveryInformation ^a	16 bits	See Table 77–7 for the internal structure of the discoveryInformation field.

^aOnly present in 10G-EPON REGISTER_REQ MAC Control indication (Clause 77).

Table 31A–6—REGISTER MAC Control indications

REGISTER (opcode 0x0005)		
indication_operand_list element	Value	Interpretation
SA	48 bit	MAC address of OLT to which registration is being performed.
ID	16 bit	LLID assigned by the OLT for use by the ONU.

Table 31A-6—REGISTER MAC Control indications (*continued*)

REGISTER (opcode 0x0005)		
indication_operand_list element	Value	Interpretation
status	accepted	Indicates that the requested registration is successful.
	denied	Indicates that the requested registration attempt is denied by the higher-layer-entity.
	deregistered	Indicates that the ONU has been deregistered, i.e., the associated LLID has been deallocated.
	reregistered	Indicates that the ONU is explicitly asked to re-register.
laserOnTime ^a	8 bits	Indicates the Laser On Time characteristic for the given ONU transmitter, expressed in the units of time_quanta.
laserOffTime ^a	8 bits	Indicates the Laser Off Time characteristic for the given ONU transmitter, expressed in the units of time_quanta.

^aOnly present in 10G-EPON REGISTER MAC Control indication (Clause 77).

Table 31A-7—REGISTER_ACK MAC Control indications

REGISTER_ACK (opcode 0x0006)		
indication_operand_list element	Value	Interpretation
SA	48 bit	MAC address of ONU to which registration is being performed.
ID	16 bit	Indicates the LLID allocated to the ONU.
status	accepted	Indicates the station has been recognized as participating in gated transmission.
	denied	Indicates the station has been excluded from participating in gated transmission.
	reset	Indicates the station has been requested to resubmit a registration request.
	deregistered	Indicates that the station will no longer be participating in gated transmission.
RTT	32 bit	Indicates the calculated round trip time for the station, as calculated following the REGISTER_ACK message reception.

Table 31A-8—EXTENSION MAC Control indications

EXTENSION (opcode 0xFFFE)		
indication_operand_list element	Value	Interpretation
OUI	24 bits	Organizationally-Unique Identifier (OUI) or Company ID (CID) that determines the format and semantics of the Value field and its subfields, if any are defined.
Value	Variable	Organization-specific value, distinguished by the OUI or CID.

Table 31A–9—PFC MAC Control indications

PFC (opcode 0x0101)		
indication_operand_list element	Value	Interpretation
priority_enable_vector	2 octets	The most significant octet is reserved (i.e., set to zero on transmission and ignored on receipt). Each bit of the least significant octet indicates if the corresponding field in the time_vector parameter is valid. The bits of the least significant octet are named e[0] (the least significant bit) to e[7] (the most significant bit). Bit e[n] refers to Priority n. For each e[n] bit set to one, the corresponding time[n] value is valid. For each e[n] bit set to zero, the corresponding time[n] value is invalid.
time_vector	a list of eight 2-octet unsigned integers	A list of eight 2-octet fields named time[0] to time[7]. The eight time[n] values are always present regardless of the value of the corresponding e[n] bit. Each time[n] field is a 2-octet, unsigned integer containing the length of time for which the receiving station is requested to inhibit transmission of data frames associated with Priority n. The field is transmitted most significant octet first, and least significant octet second. The time[n] fields are transmitted sequentially, with time[0] transmitted first and time[7] transmitted last. Each time[n] value is measured in units of pause_quanta, equal to the time required to transmit 512 bits of a frame at the data rate of the MAC. Each time[n] field can assume a value in the range of 0 to 65 535 pause_quanta.